

Changing connections: temporal potential structural
connectivity, unprotected pinch points, and monitoring
priorities for free-ranging cheetahs (*Acinonyx jubatus*)
in southern Africa

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Abstract

[TODO: Context: cheetah range contraction, majority of range outside protected areas.]

The southern African Cheetah (*Acinonyx jubatus jubatus*), has had their range decreased every since their last genetic bottleneck around 12,000 - 18,000 years ago during the last mega-faunal extinction. This has only been exacerbated in the past decades due to both natural and anthropogenic reasons, leaving their current extant range majority outside protected areas. This leaves "barriers" like ranchers, roads and other human development in areas where cheetahs will be roaming.

[TODO: Gap: existing regional work models habitat suitability, not observed change in network permeability; the 2026 KAZA multispecies assessment excluded cheetah outright.]

Models looking at habitat suitability have been done in the past, but none in regards to observed change in the movement network permeability. One 2026 KAZA (Kavango-Zambezi Transfrontier Conservation Area) multi-species connectivity study excluded the cheetah outright in their study.

[TODO: What we did: held range cores fixed at the 2010–2016 confirmed-range baseline and modelled potential structural connectivity across five predeclared resistance scenarios at four snapshots spanning 2012–2024.] [TODO: Headline result: magnitude and geography of connectivity loss.] [TODO: Second result: share of robust pinch points outside protected areas.] [TODO: Third result: ranked monitoring cells and their stability across weightings.] [TODO: Claim boundary sentence: these are modelled structural priorities, not demonstrated cheetah movement.]

Keywords: landscape connectivity; circuit theory; *Acinonyx jubatus*; protected area coverage; monitoring prioritisation; southern Africa

1 Introduction

Move 1: Why cheetah connectivity, and why outside protected areas

[TODO: Global decline and range contraction. The fact that most southern African cheetahs persist on unprotected land is the whole reason a connectivity paper is a conservation paper rather than a park-management paper. Cite the range assessment and the global decline synthesis. Keep to context; no resistance calibration here.]

The southern African cheetah has had their range substantially reduced from their original range, with their current extant range being on majority unprotected land. Cheetah range on protected land is governed with park-management initiatives and are much easier to identify conservation efforts. Unprotected land serves a different type of challenge, with many stakeholders and many variables that are not easily controllable. This paper not only looks at land change between 2020 and 2024, but also today, and provides a predictive look at what future land permeability for cheetah movement looks like based on the trends identified.

Move 2: What is already modelled, and what is not

[TODO: Habitat suitability for this species and region is now well covered, including current and projected suitability with protected-area mismatch. State plainly that this project does not add another suitability model. Then the pivot: suitability is not permeability, and no published work reports how permeability between established cheetah range cores has *changed*. Note that the recent regional multispecies connectivity assessment excluded cheetah for insufficient data, which is simultaneously the justification and the constraint.]

Move 3: The methodological frame

[TODO: Least-cost and circuit-theoretic connectivity (Adriaensen et al., 2003; McRae et al., 2008); resistance parameterisation is the dominant source of uncertainty in this class of model (Zeller et al., 2012, 2018); hence a predeclared scenario ensemble rather than a single preferred surface. Graph-level importance follows Saura and Pascual-Hortal (2007); metric choice follows Keeley et al. (2021). Design choices are framed against Beier et al. (2008).]

Move 4 — Objectives and hypotheses

[TODO: State the three objectives (change, protection gap, monitoring priority) and list H1–H4 with their directional predictions. Every hypothesis must be falsifiable as written; if you cannot describe the result that would refute it in one clause, rewrite it. See the Hypotheses sheet.]

Terminology. Throughout, potential structural connectivity refers to modelled landscape permeability between fixed range cores. It is not evidence of cheetah movement, dispersal, or gene flow. Where a location is described as a pinch point, this means a cell of high modelled current concentration, not an observed crossing.

2 Methods

2.1 Study area and analysis extent

[TODO: WF-001. Countries, area, the confirmed-range envelope, exclusion rules, and the statement that the extent was fixed before any connectivity output was viewed.] Analyses were conducted in [TODO: equal-area CRS, e.g. Africa Albers Equal Area Conic, give the full WKT in Appendix A] at a working resolution of 1 km.

2.2 Rationale for the working resolution

[TODO: CAU-006. The occurrence baseline is a 10-km assessment grid; environmental covariates range from 30 m to 1 km. Justify 1 km as the compromise: fine enough that a pinch point is a meaningful location, coarse enough that it does not imply precision the occurrence evidence cannot support. State that the full pipeline was repeated at 2 km and 5 km as a resolution sensitivity (WF-022).]

2.3 Range cores

[TODO: WF-011. Three or more core definitions from the archived confirmed-range data, differing in evidence-category threshold and minimum patch area. Report how many cores each definition yields. State explicitly that cores are HELD CONSTANT across all four temporal snapshots, and why (CAU-002): no comparable range reassessment exists after the baseline period, so any core movement would be invented.]

2.4 Covariates and their temporal treatment

[TODO: Table 1 lists every layer, its native resolution, its source year(s), and critically whether it is treated as DYNAMIC (varies across snapshots) or STATIC (held fixed, contributing no temporal signal). Do not bury this distinction. Roads, livestock, terrain and hydrography are static; built-up fraction, nighttime radiance and continuous vegetation fields are dynamic.]

Why categorical land cover is not the change carrier. [TODO: CAU-008 and the MODIS class-instability problem. Annual categorical products flip classes between years for reasons unrelated to landscape change. Temporal signal is therefore carried on continuous fields; categorical land cover is used for masking and description only.]

2.5 Resistance scenarios

[TODO: WF-013, WF-014. Present RES-01 to RES-05 as a table of variable-to-resistance functions with the evidence or rationale for each. State that the scenario set was fixed before results were viewed and that no single scenario is designated as preferred. Report the transformation (linear vs exponential) per scenario and cite Zeller et al. (2012) for why the transformation matters as much as the weights.]

2.6 Connectivity modelling

[TODO: WF-015, WF-016. Least-cost / effective-distance work in ArcGIS Pro (Distance Accumulation, Optimal Region Connections, Corridor). Current flow computed with Circuitscape.jl (Anantharaman et al., 2020) in pairwise mode; give the solver, the source/ground configuration, and how current was normalised for comparability across years. State why the omnidirectional approach (Landau et al., 2021) was NOT used: cores are defined a priori, so a core-to-core formulation is the correct one.]

Corridor definition. [TODO: Decide and state ONCE whether corridors are (a) continuous current surfaces summarised by quantile or (b) thresholded polygons. This choice sets the denominator for the protected-area comparison in §2.9 and cannot be made after seeing the overlap result.]

2.7 Network structure

[TODO: WF-017. Cores as nodes, effective distances as edge weights. Report the probability of connectivity and per-node dPC (Saura & Pascual-Hortal, 2007), computed in Conefor (Saura & Torné, 2009) or an equivalent, and betweenness on the same graph. Justify the metric set against Keeley et al. (2021). Report sensitivity to the dispersal-distance parameter; do not report a single value.]

2.8 Temporal comparison

[TODO: WF-018. Same cores, same scenarios, same resolution across all four snapshots, so that differences are attributable to covariate change alone. Give the change equations. Report both absolute and proportional change.]

2.9 Protected-area coverage

[TODO: WF-019, CAU-011, CAU-012. Name the WDPA <MONTH YEAR> release. State polygon-only handling and what was excluded. State the null explicitly: high-current area is compared against [TODO: the full extent OR the modelled network] using an area-standardised permutation test. Predeclare this.]

2.10 Robustness and uncertainty

[TODO: WF-022. The full sensitivity grid: five resistance scenarios $\times$ core definitions $\times$ resolutions $\times$ snapshots. Report the resulting run count. Define "robust" numerically (e.g. a cell in the top decile of current under at least k of n runs) and fix k in advance. Produce an agreement map. Low-agreement locations are reported as uncertain, never as priorities.]

2.11 Independent plausibility assessment

[TODO: WF-020, WF-021, CAU-004, CAU-017. See the validation decision note before writing this. If the plausibility-check route is taken, this subsection is titled as above and must NOT use the word "validation". If the empirical-resistance route is taken, add RES-06, describe the fitting procedure, and report country-held-out performance following Roberts et al. (2017).]

2.12 Monitoring prioritisation

[TODO: WF-023, WF-024, CAU-020. Components, normalisation, and the predeclared weight set, plus at least two alternative weightings. Report rank stability (Spearman / Kendall) across weightings. Publish component scores alongside the composite so a reader can reweight.]

2.13 Software and reproducibility

[TODO: ArcGIS Pro [TODO: version]; Circuitscape.jl [TODO: version]; Julia [TODO: version]; R [TODO: version] or Python [TODO: version]; Google Earth Engine for acquisition and harmonisation. All geoprocessing scripted in arcpy rather than ModelBuilder. Archive location and what can and cannot be redistributed (CAU-012, CAU-013).]

3 Results

3.1 Connectivity change, 2012–2024

[TODO: H1. Magnitude and geography of change in current flow and effective distance. Report per country and per core-pair. Figure 1.]

3.2 Robust pinch points

[TODO: H3. How many locations met the robustness criterion, and how concentrated they are. Figure 2, Table 2.]

3.3 Protected-area coverage of priority connectivity

[TODO: H2. Observed share outside protected areas versus the predeclared null, with the permutation result. Table 3.]

3.4 Model sensitivity

[TODO: Agreement across the sensitivity grid. This subsection should be substantial. If corridor rankings are unstable, say so here plainly; the Discussion then pivots to the uncertainty-guided monitoring framing rather than a corridor map. Figure 3.]

3.5 Independent plausibility assessment

[TODO: Outcome of §2.11. Report it even if weak or mixed.]

3.6 Monitoring priorities

[TODO: H4. Ranked cells and their stability. Figure 4, Table 4.]

Figure 1: Change in modelled current flow between snapshots, 2012–2024. [TODO: Colour-blind-safe diverging ramp; uncertainty shown, not hidden.]

Figure 2: Robust pinch points and their protected-area status.

Figure 3: Ensemble agreement across the sensitivity grid.

4 Discussion

What the results mean, at the level the evidence supports

[TODO: First paragraph restates the headline finding in one sentence and immediately bounds it. Resist writing "cheetah movement" anywhere in this section.]

Figure 4: Ranked monitoring priority cells.

Table 1: Covariates, native resolution, source years, and static/dynamic treatment.

4.1 Where permeability has declined, and what co-occurred

[TODO: H1. Frame as spatial co-occurrence, not causation (CAU-014). If the dominant pressure differs by country, report the heterogeneity rather than compressing it into a single anthropogenic narrative.]

4.2 The protection gap as a governance question

[TODO: H2. If the outside-PA share is high, the defensible reading is that connectivity for this species depends on land under other management, which is a statement about where conservation effort must operate, not evidence that protected areas have failed. Note the confound that protected areas are often sited in land of low agricultural value.]

4.3 Robust versus model-sensitive linkages

[TODO: H3. This is the paper’s most transferable contribution. Robust locations support action; model-sensitive locations define the survey agenda. Note the caveat that scenarios sharing input layers can agree artificially.]

4.4 From uncertainty to a survey design

[TODO: H4. The monitoring index is decision support under stated assumptions, not an objective ranking (CAU-020). Say what a partner organisation would actually do with it, and acknowledge that accessibility and cost will reorder the list in practice.]

Table 2: Top 20 robust pinch points with country, protection status, and agreement score.

Table 3: Protected-area coverage of high-current area against the predeclared null.

Table 4: Monitoring priority ranking with component scores under alternative weightings.

4.5 Comparison with regional connectivity work

[TODO: Position against the elephant and multispecies transfrontier assessments (Brennan et al., 2020; Naidoo et al., 2024) as methodological neighbours, and against the fencing-effects literature (Osipova et al., 2018). Be explicit that those are different species with different movement ecology: the methods transfer, the resistance values do not (CAU-015).]

5 Limitations

Fixed historical cores. [TODO: CAU-002. The single most important limitation. Cores derive from 2010–2016 evidence and do not move; all temporal signal is landscape-side.]

Structural, not functional. [TODO: CAU-001.]

Sampling bias in the occurrence baseline. [TODO: CAU-003. Absence of records is not evidence of absence.]

Resistance is expert-assigned. [TODO: CAU-005. What the scenario ensemble does and does not compensate for.]

Static covariates. [TODO: Roads, livestock, fences and terrain contribute no temporal signal; consequences for interpreting change.]

Mapping completeness varies by country. [TODO: CAU-010.]

Geographic transferability. [TODO: CAU-015.]

Scope. [TODO: CAU-018. Climate, conflict and occupancy are deliberately out.]

6 Conclusion

[TODO: Three or four sentences. One on what changed, one on where the robust priorities are and their protection status, one on the monitoring recommendation, one closing the claim boundary. No new information, no hedging language you have not already earned.]

Data availability

[TODO: CAU-012, CAU-013. Derived outputs and all code are archived at [TODO: OSF/Zenodo DOI]. Raw protected-area data are not redistributed and must be obtained from Protected Planet under its terms. Occurrence data are shared only at the resolution permitted by the source archive.]

Acknowledgements

[TODO: Department, any data providers, any partner organisation that reviewed the monitoring priorities.]

Conflicts of interest

The author declares no conflicts of interest.

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A Analysis protocol as predeclared

[TODO: Paste the locked protocol document here verbatim, with its version and date. This is what lets you write "predeclared" in the Methods without overclaiming. It should include: extent and exclusion rules, full CRS WKT, working resolution and the alternatives to be tested, core definitions, the resistance scenario table, the sensitivity grid, the robustness threshold k , the MCDA weight sets, and the claim dictionary listing prohibited phrasings.]

B Supplementary material

[TODO: Temporal alignment table (CAU-007); correlation/VIF screening among human pressure covariates (CAU-016); per-scenario current maps; per-resolution results; full monitoring ranking with component scores; software environment manifest.]